Predicting Next-Best-Product Adoption in Retail Banking from the First 90 Days of Transactions

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ANA540 Machine Learning, Final Project

October 2026

Abstract

We predict loan and credit-card adoption during the year following a 90-day transaction window using 4,303 accounts from the Berka data. A prior baseline, elastic-net logistic regression, LightGBM, product-specific Cox models, random survival forests (RSF), an RMTPP-style GRU and an ensemble are compared using earlier opening cohorts for training and the 1997 cohort for testing. Hyperparameters, features and decision weights are selected on expanding-window validation folds. Adoption prevalence rises from 12.5% in training to 23.5% in testing. Tuned models achieve macro AUCs of 0.77–0.82 but predict the majority class under arg-max. A learned decision rule raises macro-F1 from 0.217 to 0.33–0.43, at the cost of accuracy. RSF achieves the highest test macro AUC (0.821) and macro-F1 (0.432); the cross-validation-selected ensemble achieves the highest loan average precision (0.402). The sequence model does not outperform aggregate features. Results support ranking potential, subject to small positive classes, cohort shift and training-label availability after some test prediction dates.

Keywords: next-best-product, customer behavior, imbalanced classification, survival analysis, temporal validation, banking

1 Introduction

Can a customer's first 90 days of transactions predict whether they adopt a loan, a credit card, both or neither during the following year? This task combines rare outcomes, censored adoption times and changing adoption rates across account-opening cohorts. Transaction histories may add behavioral information beyond demographic snapshots, but repayments and card activity after adoption would reveal the outcome.

We therefore compare aggregated-feature, survival and sequence models under a fixed landmark and horizon. The study separates the contributions of hyperparameter tuning, feature selection and a learned decision rule, using opening-year validation folds and evaluation on a held-out cohort. The aim is to predict observed adoption; the data do not identify responses to marketing offers.

2 Related Work

Time to adoption and landmarking. Adoption of a financial product can be modelled as a time-to-event outcome. The Cox proportional-hazards model [1] and random survival forests [2] handle right censoring and are evaluated by the concordance index [3]. Summarizing a customer's longitudinal transaction history up to a fixed observation point (day 90) to predict subsequent adoption within a fixed horizon follows the survival *landmarking* framework [4, 5], which has recently been combined with machine learning ensembles for dynamic risk prediction [6, 7]. When several events are possible, cause-specific or subdistribution hazards [8] are standard; here loans and cards are not mutually exclusive competing events, so each product receives its own cause-specific hazard model.

Neural event models and banking streams.

Recurrent marked temporal point processes (RMTPP) [9] embed timestamped, typed events with a recurrent network (such as a GRU [10]) to parameterize the conditional intensity of future arrivals [11]. In retail banking, recent work has applied sequence encoders and deep representation learning to raw transaction streams for cash-flow analysis, change-point detection, and next-product recommendation [12–14]. We adapt the intensity-plus-marks paradigm to predict product adoption *after* a 90-day landmark rather than the next transaction inside the stream.

Tabular learners vs. deep models. Regularized linear models [15] and gradient-boosted trees [16] remain strong baselines for structured financial data. Extensive recent benchmarks demonstrate that tree-based ensembles consistently outperform deep neural networks on medium-sized tabular datasets ($N < 10{,}000$) due to skewed feature distributions, irregular decision boundaries, and rotational non-invariance [17, 18], motivating a direct comparison between hand-engineered 90-day aggregates and raw event sequences.

Imbalance, thresholds and shift. Under strong class imbalance, precision-recall curves are more informative than accuracy or ROC-AUC [19]. Hard classifications require an explicit decision rule rather than default arg-max, using cost-sensitive [20] or F1-optimal thresholds [21], while shifts in class priors between training and deployment cohorts require probability or threshold adjustment [22].

Validation under time. When observations are ordered chronologically, random splits overstate generalization, requiring expanding-window validation and out-of-time cohort testing [23, 24]. Hyperparameters are tuned via random or grid search [25], and test uncertainty is quantified via the bootstrap [26].

3 Data

3.1 The Berka data set

We use the Berka data set [27], a real, anonymized extract from a Czech bank released for the PKDD'99 Discovery Challenge. Its eight tables describe 4,500 accounts, 5,369 clients, 77 districts, 682 loans, 892 credit cards, 6,471 standing orders and 1,056,320 transactions between January 1993 and December 1998. It combines exact transaction dates and amounts for every account with dated loan and card records, which allows adoption to be treated as an event in time observed against the customer's behavior history. Each account holds at most one loan and one card. A loan or card is the outcome of the bank's approval as well as the customer's interest and there are no marketing-contact data, so we model *adoption*, not response to an offer.

3.2 Exploratory findings

Placeholders in the raw tables ("?" and blanks) were recoded as nulls, giving 2,210,119 null cells, almost all of them structural (e.g., `operation` is blank exactly for interest credits, and the counterparty fields are filled only for transfers). The only genuine gaps are two district fields for one district (48 accounts). Amounts and balances are heavily right-skewed; 2,999 negative-balance transactions on 288 accounts are genuine overdrafts.

Measured at the end of the data, 68.8% of accounts hold neither product, 16.0% a card only, 11.4% a loan only and 3.8% (170 accounts) both. Timing is highly structured (Fig. 1): no account adopts before day 102 after opening; loans arrive a median of 395 days after opening and never after day 697, so loan incidence plateaus at about 16%; cards start after day 203, arrive later (median 658 days) and keep accumulating (26.9% by five years). Card issuance also rises steeply with calendar time (1 card in 1993, 449 in 1998), so card adoption mixes account age with the bank's roll-out.

Early behavior is informative. Over the first 90 days, eventual adopters have more than twice the median deposits of *neither* accounts (66–85k vs. 31k CZK), a much higher typical balance (32–39k vs. 19k CZK), a larger cash-withdrawal share (22–30% vs. 12%), and almost never receive a pension (0% of *loan only* and *both*, 2.5% of *card only*, vs. 23.3% of *neither*). Age matters (87% of customers aged 55+ end with neither product vs. 62–66% in younger bands), whereas gender, region and district-

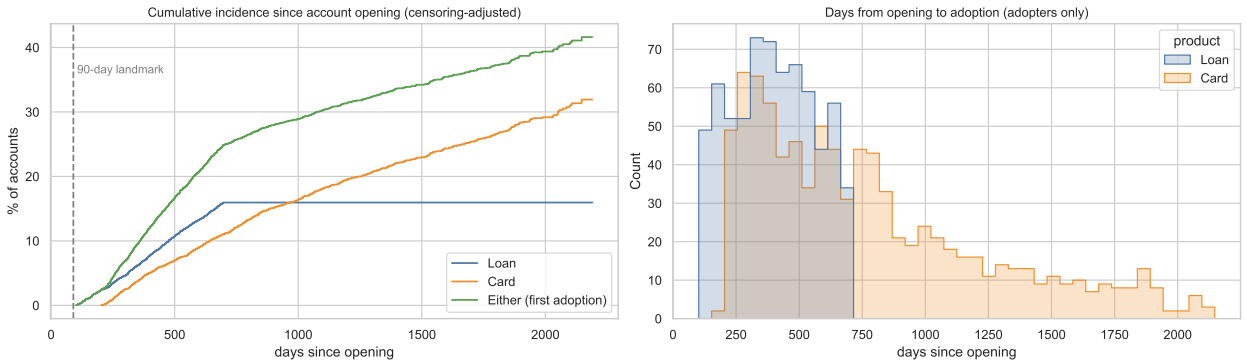


Figure 1. Timing of adoption. Left: censoring-adjusted cumulative incidence since account opening (dashed line: 90-day landmark). Right: days from opening to adoption for adopters. Loans plateau after day 697, cards keep accumulating.

level context hardly do (29–33% adopt in every salary, unemployment and urbanization quartile).

Leakage. Loan repayments (13,580 transactions with purpose `UVER`) and card withdrawals (8,036 transactions) occur only on or after the adoption date. Using the full transaction history would therefore reveal the label, which motivates a *landmark* design with predictors restricted to the first 90 days.

4 Problem Formulation and Preprocessing

4.1 Landmark, horizon and target

Predictors contain transactions from days 0–90 and static account/customer attributes. The target is adoption during $H = 365$ days after the day-90 landmark, with classes *neither*, *loan only*, *card only* and *both*. This equal follow-up window avoids the unequal exposure of an end-of-data label (Fig. 2). Accounts with less than 365 days of post-landmark follow-up are excluded (197 accounts), leaving 4,303 labelled accounts.

For survival training, each product additionally has a time measured from the landmark to its adoption or administrative censoring on 31 December 1998. Adoption of the other product does not end follow-up. These targets can extend beyond the classification horizon.

Table 1. Class distribution of the labelled accounts (adoption within 365 days of the landmark).

Class	Train	Test	Total
neither	3,150 (87.5%)	536 (76.5%)	3,686
loan only	306 (8.5%)	69 (9.8%)	375
card only	127 (3.5%)	82 (11.7%)	209
both	19 (0.5%)	14 (2.0%)	33
Accounts	3,602	701	4,303
Any product	12.5%	23.5%	14.3%

4.2 Time-based splits

Accounts opened before 1997 form the training set (3,602 accounts) and those opened in 1997 form the test set (701 accounts), which mimics scoring new customers with a model built on the past. Table 1 shows the class distribution: the 1997 cohort contains far fewer *neither* accounts (76.5% vs. 87.5%) and a much higher share of cards, so class priors shift between training and test. Inside the training set, tuning uses two *expanding-window* folds: train on 1993–94 and validate on 1995 (1,578/661 accounts), and train on 1993–95 and validate on 1996 (2,239/1,363 accounts). The first training fold contains a single *both* account and the second eight, which limits what can be learned about *both* during tuning.

4.3 Features and preprocessing

The 46 raw predictors summarize transaction counts and amounts, balances, inter-transaction gaps, re-

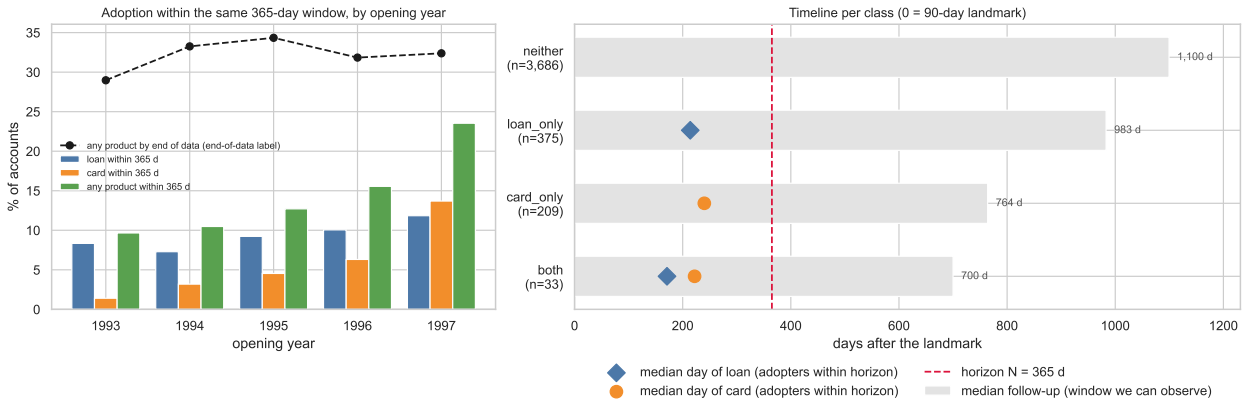


Figure 2. Left: adoption within the same 365-day window by opening year rises steeply (any product 9.7% for 1993 to 23.5% for 1997), whereas the end-of-data rate (dashed) stays near 29–34% because older accounts had more time. Right: median observable window per class (grey), the 365-day horizon (red line) and median adoption day of adopters inside the horizon.

cency, cash/transfer shares, payment-purpose indicators and static attributes. Preprocessing uses median imputation, signed-log transforms for 21 skewed variables, one-hot encoding and redundancy pruning (Spearman $|\rho| > 0.90$). Fitted steps use only the training portion of each fold. The final *base* set has 41 encoded inputs; linear and Cox models use standardized inputs, while trees use the unscaled representation.

Engineered features. Ten additional features describe average deposits/payments, transactions per active day, balance change/range, incoming-transaction share, inflow relative to balance and district salary, cash intensity, and maximum-to-median amount. The *eng* set has 51 inputs.

Event sequences. For the neural model, each account's 90-day transaction stream is kept as a sequence of events with the log gap since the previous event, log amount, time within the window and three categorical marks (type, operation, purpose).

5 Methods

5.1 Models

All methods produce probabilities for the same four horizon outcomes. Randomized fits use seed 42 unless otherwise stated; selected settings appear in Table 2.

Prior baseline. Each account receives the training class proportions, providing a no-feature benchmark under imbalance.

Elastic-net logistic regression. A multinomial softmax model with combined ℓ_1 and ℓ_2 penalties [15] provides a regularized linear baseline on scaled aggregate features. It tests whether additive effects suffice.

LightGBM. Multiclass gradient-boosted trees [16] use unscaled aggregate features to capture nonlinear relationships and interactions without specifying them in advance.

Product-specific Cox regression. Separate ridge-penalized Cox models [1] estimate loan and card hazards, $h_p(t \mid x) = h_{0p}(t)\exp(x^\top \beta_p)$. They incorporate censoring while assuming proportional hazards within each product.

Random survival forest (RSF). Separate forests [2] estimate each product's survival function from aggregate features, allowing nonlinear effects and interactions without proportional hazards. Cox and RSF use `lifelines` and `scikit-survival`, respectively [28, 29].

Shared survival-to-class conversion. For either survival method, the predicted probabilities of adopting a loan and a card within the horizon, p_{loan} and p_{card}, are combined assuming conditional

independence of loan and card adoption:

$$\begin{aligned} P(\text{neither}) &= (1-p_{\text{loan}})(1-p_{\text{card}}),\\ P(\text{loan only}) &= p_{\text{loan}}(1-p_{\text{card}}),\\ P(\text{card only}) &= (1-p_{\text{loan}})p_{\text{card}},\\ P(\text{both}) &= p_{\text{loan}}p_{\text{card}}. \end{aligned} \tag{1}$$

This approximation does not explicitly model residual dependence between products; its direction of error is not determined a priori.

RMTPP-style GRU. A GRU [10] tests whether event order adds information beyond aggregates. Continuous event inputs and 4-dimensional embeddings of type, operation and purpose feed the recurrent encoder. Its final state is concatenated with static attributes and passed through a shared 64-unit ReLU layer with dropout. Two heads output four-class softmax probabilities and product-specific hazards $\log \lambda_{ip}(t) = a_p(z_i) + w_p t$, where z_i is the shared representation and t is years since the landmark. Thus the prediction horizon is $t = 1$. This adapts RMTPP's exponential intensity parameterization [9] to post-landmark adoption, rather than next-transaction prediction.

The classification and survival heads are trained jointly:

$$\begin{aligned} \mathscr{L} &= \mathscr{L}_{\text{CE}} + \alpha \mathscr{L}_{\text{surv}},\\ \mathscr{L}_{\text{surv}} &= \frac{1}{N}\sum_{i=1}^{N}\sum_{p\in\{\ell,c\}} \big[\Lambda_{ip}(T_{ip}) \\ &\qquad - \delta_{ip}\log \lambda_{ip}(T_{ip})\big], \end{aligned} \tag{2}$$

where T_{ip} is event or censoring time, δ_{ip} indicates observed adoption, $\Lambda_{ip}(t) = \int_0^t \lambda_{ip}(u)\,du$, and α controls the survival-loss contribution. Intensity-head probabilities are $1-\exp[-\Lambda_{ip}(1)]$, combined by Eq. 1. The direct softmax head does not impose that independence approximation.

PyTorch [30] training uses Adam, batches of 128, at most 80 epochs and early stopping after 10 epochs without sufficient improvement in the joint validation loss. A random stratified 15% inner holdout is drawn from each training portion (singleton classes are merged for stratification only); outer model selection remains chronological. Cross-validation selects both the configuration and output head; the reported selected GRU uses the classification head. The inner holdout remains excluded from gradient updates in the final fit.

Ensemble. Equal-weight averaging of the five tuned models' four-class probability vectors tests whether their predictions complement one another. Its decision rule is tuned separately after averaging.

Supervision differs across families. Cox, RSF and the GRU survival loss use event/censoring information beyond day 365, whereas the classifiers use only horizon outcomes. The comparison therefore varies both architecture and training-outcome information.

5.2 Optimization

Validation objective. On each expanding-window fold, the search score is the mean of loan and card average precision (AP, reported as PR-AUC), computed using $p_{\text{loan}} = P(\text{loan only}) + P(\text{both})$ and $p_{\text{card}} = P(\text{card only}) + P(\text{both})$. The two fold scores are averaged equally. This avoids selecting on nearly constant arg-max macro-F1; loan and card ROC-AUC are also reported.

Hyperparameter tuning. Table 2 lists the search spaces. ElasticNet, Cox and RSF use exhaustive search; LightGBM uses 30 random configurations. The GRU evaluates its default plus 10 sampled configurations, each with both heads [25]. Initial configurations are scored on the same folds.

textbfFeature engineering and selection. With the best hyperparameters of each model fixed, six feature sets are compared on the same folds: *base*; *eng*; *top-k* ($k = 10, 15, 20, 30$ highest-ranked columns of base plus engineered features by LightGBM gain, ranked on each fold's training accounts); and, for ElasticNet only, the first n principal components of the scaled base set ($n = 5, 10, 15, 20$). A smaller set is preferred whenever it scores within 0.005 of the best. The GRU keeps its own input and is not part of this comparison.

Decision rule. The best configuration of every model is re-run on both folds to obtain out-of-fold (OOF) probabilities for all 2,024 validation accounts. The four class probabilities are multiplied by $[1, w_\ell, w_c, w_\ell w_c]$ before the arg-max, and (w_ℓ, w_c) are chosen on a logarithmic grid (48 values

Table 2. Search spaces and selected configurations (selected values in the last column).

Model	Search space	Selected
ElasticNet	$C \in \{0.03, 0.1, 0.3, 1, 3\}$; ℓ_1 ratio $\in \{0.15, 0.5, 0.85\}$; class weight $\in$ {none, balanced} (30 configs)	$C = 0.3$, ℓ_1 ratio 0.85, none; base (41)
LightGBM	leaves $\{4, 8, 15, 31\}$; learning rate $\{0.01, 0.03, 0.05, 0.1\}$; trees $\{100, 200, 400\}$; min. child samples $\{10, 20, 40, 80\}$; ℓ_2 $\{0, 1, 5, 20\}$; column sample $\{0.5, 0.7, 1\}$; row sample $\{0.7, 1\}$; class weight (30 random configs)	31 leaves, rate 0.01, 100 trees, min. child 80, no regularization, no sampling; eng (51)
Cox	ridge penalty $\{0.01, 0.03, 0.1, 0.3, 1, 3\}$	0.01; eng (51)
RSF	min. samples per leaf $\{5, 15, 30, 60\}$; max. features $\{\sqrt{p}, 0.5p\}$; 200 trees (8 configs)	leaf 15, $0.5p$; top-15
GRU	hidden $\{16, 32, 64\}$; dropout $\{0.1, 0.2, 0.4\}$; weight decay $\{10^{-5}, 10^{-4}, 10^{-3}\}$; learning rate $\{10^{-3}, 3 \cdot 10^{-3}\}$; $\alpha \in \{0.5, 1, 2\}$; class-weight power $\{0, 0.5\}$; head $\in$ {class, intensity} (11 configs)	hidden 32, dropout 0.4, decay 10^{-5}, rate 10^{-3}, $\alpha = 1$, no class weights; class head

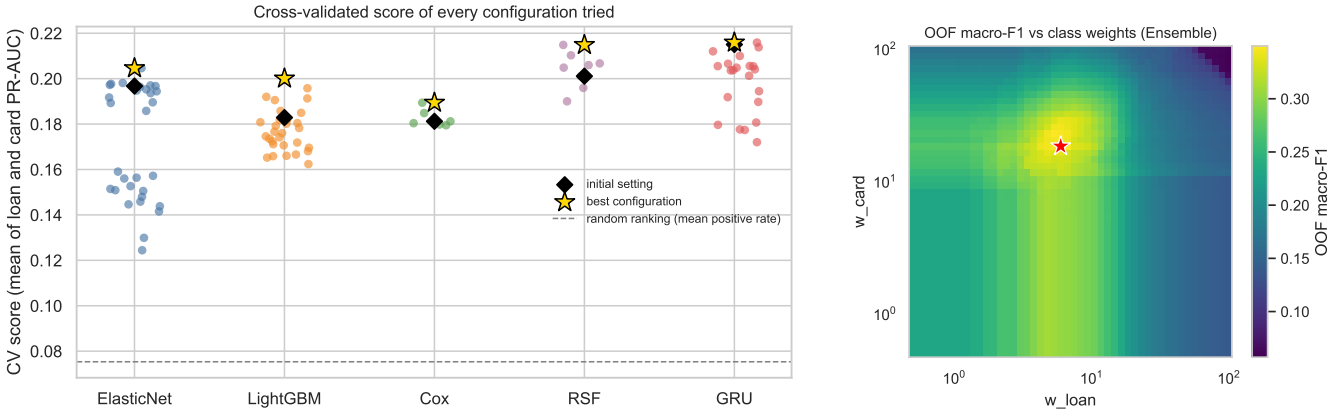


Figure 3. Optimization results on the validation folds. Left: cross-validated score (mean of loan and card PR-AUC) of every configuration tried per model; diamonds mark the untuned setting of the baseline stage and stars the best configuration. Right: validation macro-F1 as a function of the two class weights for the ensemble (red star: selected weights).

from 0.5 to 100) to maximize OOF macro-F1; ties are broken toward weights closest to one. Tying $w_{\text{both}} = w_\ell w_c$ is consistent with Eq. (1) and avoids fitting a weight for a class with only a handful of validation examples. It is a regularizing constraint, not a statement about costs, which in a real cross-selling campaign need not factorize in this way. The test set is excluded from all tuning.

5.3 Evaluation

Evaluation reports accuracy, macro precision/recall/F1, one-vs-rest ROC-AUC, per-class recall, product-level ROC-AUC and AP, and Harrell's C-index [3] for survival outputs. We use 1,000 account-level bootstrap resamples for 95% percentile intervals [26]. Three variants are reported per model: *initial* (untuned, arg-max), *tuned* (tuned hyperparameters and features, arg-max) and *tuned + rule* (additionally, the decision weights).

6 Results

6.1 Cross-validated tuning

Table 3 summarizes the validation results and Fig. 3 shows every configuration tried. Gains from tuning

Table 3. Cross-validated results (two expanding-window folds). CV score = mean of loan and card PR-AUC; ROC-AUCs are for the tuned model. OOF macro-F1 is shown for the tuned model with arg-max and with the decision rule; the latter is tuned on the same OOF predictions and therefore optimistic.

	CV score		Loan	Card	OOF macro-F1		Weights		Feature set
Model	initial	tuned	AUC	AUC	arg-max	+ rule	w_ℓ	w_c	(chosen)
ElasticNet	0.197	0.205	0.772	0.752	0.230	0.337	4.3	13.1	base (41)
LightGBM	0.183	0.210	0.768	0.774	0.230	0.348	6.7	23.1	eng (51)
Cox	0.181	0.196	0.778	0.762	0.238	0.356	7.5	18.4	eng (51)
RSF	0.201	0.218	0.778	0.785	0.230	0.350	3.0	11.7	top-15
GRU	0.215	0.216	0.793	0.745	0.236	0.329	5.3	20.6	sequence + static
Ensemble	–	0.219	0.790	0.790	0.230	0.350	6.0	18.4	—

are modest and differ by model: the CV score rises from 0.183 to 0.210 for LightGBM, 0.181 to 0.196 for Cox, 0.201 to 0.218 for RSF and 0.197 to 0.205 for ElasticNet, while the GRU's default is already close to its best (0.215 vs. 0.216). The ensemble has the highest CV score (0.219), narrowly ahead of RSF (0.218) and GRU (0.216); two folds provide limited evidence for separating these candidates.

Feature sets. Engineered features improve LightGBM, Cox and RSF, while ElasticNet prefers base features. Selected sets are listed in Table 2. Top-15 RSF retains essentially the full engineered-set score (0.2177 vs. 0.2180), satisfying the parsimony rule.

Decision rule. Selected weights are larger for cards than loans (Table 3). The ensemble optimum lies inside the search grid (Fig. 3); the rule raises OOF macro-F1 to 0.33–0.36, subject to the selection optimism noted above.

6.2 Test-set performance

Table 4 reports the held-out 1997 results and bootstrap intervals.

Ranking versus hard labels. Four initial models predict almost exclusively *neither*; LightGBM detects a few adopters. After tuning, all models have arg-max macro-F1 of 0.217 despite macro AUCs of 0.77–0.82. The prior baseline has accuracy 0.765 and AUC 0.5. Useful ranking therefore does not translate automatically into useful hard decisions.

Ranking quality. RSF has the highest test macro AUC (0.821) and card AP (0.330); the ensemble has the highest loan AP (0.402). Product base rates are 0.118 for loans and 0.137 for cards. Survival discrimination is also strongest for RSF, while Cox improves with tuning (Table 6).

Decision trade-off. The rule raises macro-F1 to 0.334–0.432 and detects adopter classes (Table 5), but accuracy falls to 0.601–0.709. The ensemble identifies 7 of 14 *both* accounts; across models, precision for *both* is only 0.07–0.14. Campaign costs are needed to judge this trade-off.

6.3 Model comparison and selection

The ensemble remains the CV-selected model even though RSF has higher test macro-F1 and AUC. The small CV margin and uncertainty in test estimates caution against declaring a universal winner.

The GRU has competitive loan ranking but weaker card ranking (Fig. 4) and greater seed variability: across five seeds, macro-F1 is 0.370 ± 0.028 for GRU and 0.422 ± 0.009 for RSF (mean $\pm$ s.d.), while LightGBM is unchanged at 0.399. Short histories (median nine events per account) and limited training size are plausible, untested explanations for the lack of a sequence-model advantage.

6.4 Calibration and prior shift

Predicted adoption probabilities are generally too low, particularly for cards (Fig. 5), consistent with the higher prevalence in 1997. Decision weights optimized on earlier cohorts do not recalibrate probabilities. AP and macro-F1 depend on prevalence; ROC-AUC is invariant only under pure prior shift

Table 4. Test-set results (1997 cohort, 701 accounts). Macro-F1 and macro AUC with 95% bootstrap intervals. "Initial": untuned, arg-max; "tuned + rule": tuned model with the decision weights of Table 3. The tuned models with plain arg-max have macro-F1 0.217 (see text).

Model	Variant	Accuracy	Macro-F1 [95% CI]	Macro AUC [95% CI]	Loan PR-AUC	Card PR-AUC
Prior baseline	initial	0.765	0.217 [0.212, 0.221]	0.500 [0.500, 0.500]	0.118	0.137
ElasticNet	initial	0.763	0.216 [0.212, 0.221]	0.780 [0.745, 0.812]	0.353	0.278
	tuned + rule	0.698	0.377 [0.322, 0.429]	0.780 [0.746, 0.812]	0.354	0.272
LightGBM	initial	0.763	0.254 [0.228, 0.283]	0.763 [0.726, 0.798]	0.300	0.275
	tuned + rule	0.625	0.399 [0.357, 0.444]	0.790 [0.758, 0.820]	0.384	0.324
Cox	initial	0.765	0.217 [0.212, 0.221]	0.777 [0.745, 0.805]	0.282	0.281
	tuned + rule	0.601	0.383 [0.341, 0.421]	0.802 [0.775, 0.827]	0.354	0.310
RSF	initial	0.765	0.217 [0.212, 0.221]	0.809 [0.783, 0.834]	0.393	0.300
	tuned + rule	0.709	**0.432** [0.380, 0.482]	**0.821** [0.794, 0.845]	0.399	**0.330**
GRU	initial	0.765	0.217 [0.212, 0.221]	0.762 [0.730, 0.794]	0.346	0.236
	tuned + rule	0.623	0.334 [0.291, 0.377]	0.767 [0.737, 0.797]	0.349	0.223
Ensemble	tuned + rule	0.643	0.404 [0.361, 0.448]	0.805 [0.779, 0.831]	**0.402**	0.303

Table 5. Test-set recall per class, initial vs. tuned + rule (the *both* class has 14 accounts).

Model	Variant	*neither*	*loan*	*card*	*both*
Prior	initial	1.00	0	0	0
ElasticNet	initial	1.00	0	0	0
	+ rule	0.84	0.20	0.22	0.36
LightGBM	initial	0.99	0.07	0.01	0
	+ rule	0.70	0.36	0.42	0.43
Cox	initial	1.00	0	0	0
	+ rule	0.67	0.38	0.33	0.57
RSF	initial	1.00	0	0	0
	+ rule	0.82	0.36	0.39	0.21
GRU	initial	1.00	0	0	0
	+ rule	0.74	0.13	0.32	0.36
Ensemble	+ rule	0.73	0.33	0.37	0.50

with unchanged class-conditional score distributions. Thus none of these results establishes performance under arbitrary cohort shift. Probability calibration and explicit prior-shift adjustment [22] remain untested.

6.5 What drives the predictions?

LightGBM relies most on balance range (17.3% of gain; 0.070 permutation drop in macro AUC) and maximum amount (16.7%; 0.061), followed by cash-related features (Fig. 6). Three of its six

Table 6. Survival C-index on the test set (loan / card): initial vs. tuned.

	Initial		Tuned	
Model	Loan	Card	Loan	Card
Cox	0.747	0.746	0.779	0.757
RSF	0.791	0.766	0.803	0.766
GRU	0.794	0.706	0.793	0.699

leading features are engineered. ElasticNet associates pensions and longer inactivity with *neither*, while larger maximum amounts point away from that class. These patterns may identify active, well-funded customers rather than credit need; they are predictive associations, not causal effects.

7 Discussion and Limitations

Small samples and selection uncertainty. The test cohort contains 83 loan adopters, 96 card adopters and only 14 *both* accounts. Two validation folds and sparse positive classes make model and decision-weight selection noisy. Test-account bootstrap intervals condition on fitted models; they do not capture all training and selection uncertainty. Seed analyses address only part of this variation.

Temporal availability and shift. Training labels and survival times extend into December 1998,

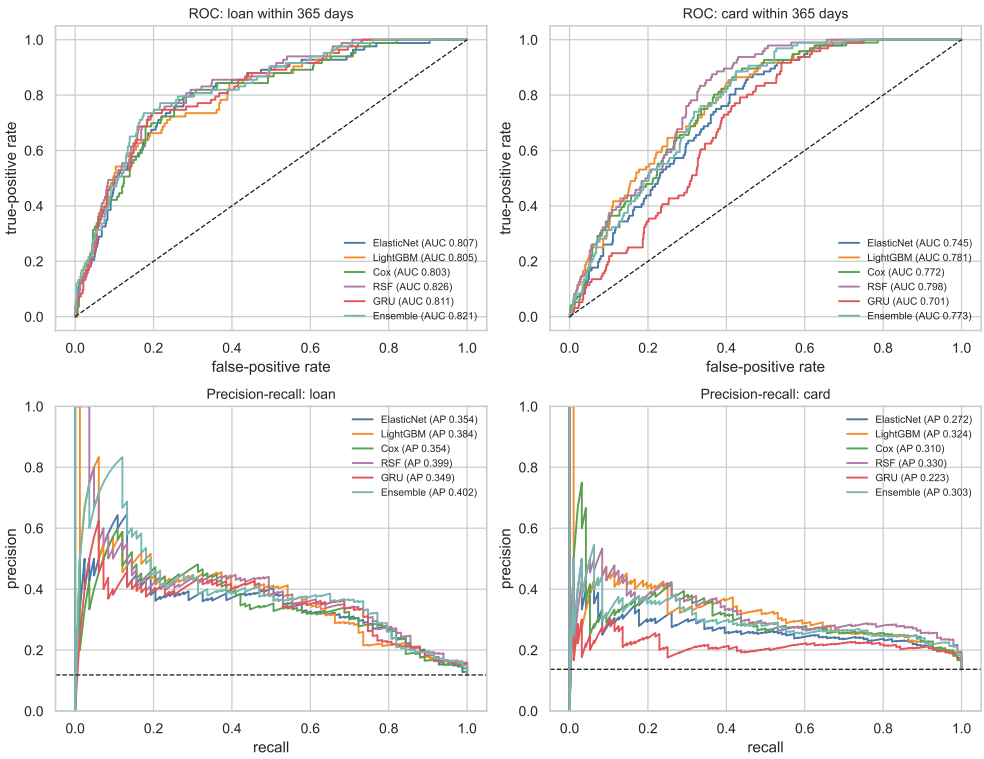


Figure 4. ROC (top) and precision-recall (bottom) curves for loan (left) and card (right) adoption within 365 days on the test set; dashed lines show random ranking.

beyond some test landmark dates. Although test labels are excluded from fitting and selection, this is a material limitation for prospective deployment. A deployment-faithful backtest should admit only labels available at each training cutoff. Higher test adoption prevalence also limits transfer of probabilities and decision weights across cohorts.

Model assumptions and supervision. Product-specific survival outputs assume conditional independence when converted to four classes. Residual dependence was not estimated. Survival models and the GRU survival loss also receive follow-up beyond the classifiers' 365-day target, so the comparison does not isolate architecture alone.

Scope and operational value. One Czech bank in the 1990s may not represent other banks or periods. Adoption reflects product availability and approval as well as customer demand. Without offer histories or campaign costs, the study cannot estimate marketing uplift or demonstrate that maximizing macro-F1 improves campaign value.

8 Conclusion

The first 90 days of transactions provide useful ranking information for subsequent loan and card adoption. Aggregate-feature models remain competitive with the GRU, and most improvement in hard-label metrics comes from the decision rule rather than hyperparameter tuning. RSF has the strongest test macro-F1 and AUC, while the CV-selected ensemble leads on loan AP. These results favor aggregate-feature tree or survival baselines for further evaluation, without establishing a definitive model winner.

Next steps are deployment-faithful temporal validation across more cohorts or banks, probability calibration, cost-based decisions and joint product modelling. Larger transaction corpora could also test whether sequence pretraining overcomes the limited histories available here.

Reproducibility. All results come from a single notebook (random seed 42; Python 3.11, scikit-learn 1.9, LightGBM 4.7, lifelines 0.30, scikit-survival 0.28, PyTorch 2.2). Figures and tables are generated by the notebook.

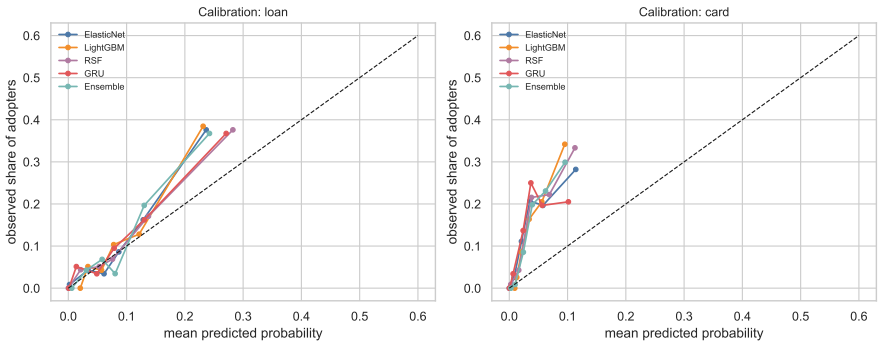


Figure 5. Calibration of the loan and card probabilities on the test set (six quantile bins).

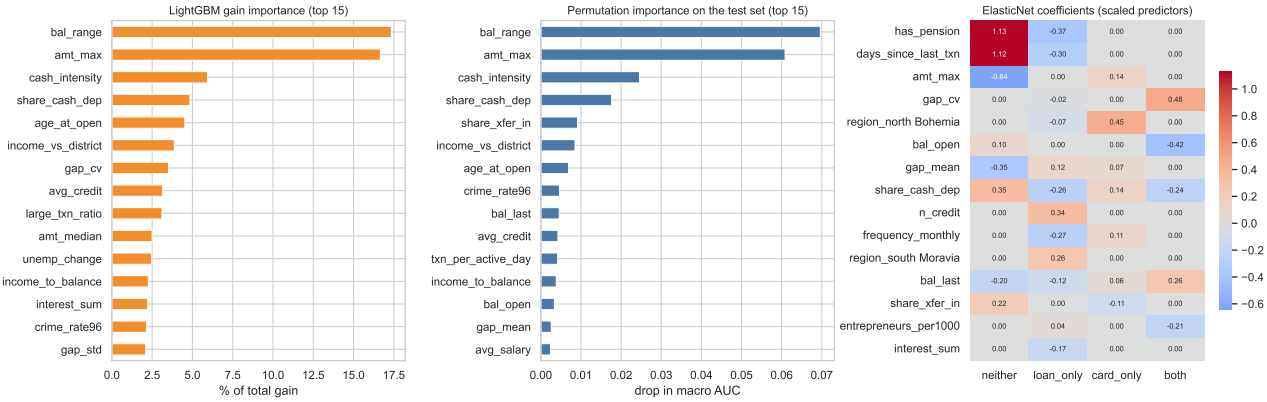


Figure 6. Left: LightGBM gain importance. Middle: permutation importance on the test set (drop in macro AUC). Right: ElasticNet coefficients (scaled predictors) for the 15 strongest effects.

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